

Bubble fluctuations

X18 (2018) — English translation

Consider an air bubble of radius R immersed in water of density ρ . At equilibrium, the pressure inside and outside the bubble is p_0 , if the surface tension of water is not taken into account. Consider air to be an ideal diatomic gas. After a slight disturbance, the bubble begins to oscillate radially with a certain frequency. The following steps will help you find this frequency. This can be done by representing the potential energy of an oscillating bubble as Ax^2 , and the kinetic energy as $B(dx/dt)^2$, where x is a small change in radius. Then the frequency will be $\omega = \sqrt{A/B}$.

A1	Find the restoring pressure force acting on a small area dS of the bubble surface when its radius changes adiabatically from R to $R + x$, ($x \ll R$).	0.70
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A2	Find the energy required to change x from 0 to some small value x_0 .	1.00
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A3	The radial speed of water at the surface of the bubble is dx/dt . Given that water is incompressible, find the radial speed of water at a distance r ($r > R$) from the center of the bubble.	1.00
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A4	Find the total kinetic energy of water.	1.00
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A5	If we neglect the air mass, the kinetic energy found in the previous paragraph will be the kinetic energy of the oscillating bubble. Find the vibration frequency.	0.50
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Now let's take into account the influence of surface tension. The surface tension coefficient σ is defined as the energy required to create a surface of unit area ($W = \sigma S$).

B1	Find the contribution of surface tension to the energy required to create a bubble of radius R in water.	0.20
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B2	Find the frequency of bubble oscillations, taking into account the influence of surface tension of water. Consider the pressure outside the bubble equal to p_0 .	1.60
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Source: <https://pho.rs/p/311>